Repeated integrals and smooth endpoint data at every order

Research proof and Lean formalization
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Abstract

We prove the all-order endpoint-data conjecture of Burke, Haris and Madhavendra. The Stieltjes measure of a continuous nonconstant increasing highest derivative is atomless and nonzero. Its finitely many moments have a strictly positive smooth representing density equal to one near both endpoints. A direct separation argument proves this representation, and repeated integration supplies the required endpoint derivatives. The admissible endpoint-data set is an invertible linear image of an open convex smooth-moment image. The full all-order source statement is formally verified in Lean, including the concrete convexity and openness proofs and every source, measure, kernel and endpoint bridge.

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1 The source conjecture

For $n \ge 0$, let $W_n \subset \mathbb{R}^{n+1}$ consist of the vectors $b = (b_0, \dots, b_n)$ for which some $f \in C^n[0,1]$ satisfies

$$f^{(j)}(0) = 0, \qquad f^{(j)}(1) = b_j \quad (0 \le j \le n), \qquad f^{(n)} \text{ is nondecreasing and nonconstant.} \tag{1}$$

Endpoint derivatives are one-sided. Burke, Haris and Madhavendra [1], Conjecture 1.2, p. 2, ask whether the following statement (P_n) holds for every n. Their Theorem A proves it for $n \le 3$.

Theorem 1. *For every integer $n \ge 0$, the set W_n is open. Every $b \in W_n$ has a witnessing $F \in C^\infty(\mathbb{R})$ such that*

$$F^{(n+1)}(x) > 0 \ (0 < x < 1), \qquad F^{(n+1)}(0) = F^{(n+1)}(1) = 1, \qquad F^{(j)}(0) = F^{(j)}(1) = 0 \ (j > n+1).$$

In fact $F^{(n+1)}$ can be chosen identically one on neighborhoods of both endpoints.

The source's sequel [2], Theorem 4.2, retains the corresponding all-order endpoint statement as a hypothesis. We prove (P_n) itself; we do not claim a separate discovery for its conditional applications or for the previously proved cases $n \le 3$.

2 The smooth moment image

Put $p_n(t) = (1, t, \dots, t^n)$, and write $m(\mu) = \int p_n \, d\mu$ for the moment vector of a finite positive measure supported on $[0,1]$. Let

$$\mathcal{A}_n = \left\{ \int_0^1 p_n(t)\rho(t)\,dt : \begin{array}{l} \rho \in C^\infty(\mathbb{R}),\ \rho > 0 \text{ on } [0,1], \\ \rho = 1 \text{ near 0 and near 1} \end{array} \right\}. \tag{2}$$

All neighborhoods in this definition are neighborhoods in $\mathbb{R}$.

Lemma 2. *$\mathcal{A}_n$ is nonempty, convex and open, and $0 \in \overline{\mathcal{A}_n}$.*

Proof. The density one proves nonemptiness. Convex combinations preserve positivity and both endpoint germs. To prove openness at a vector represented by ρ, fix a nonnegative nonzero smooth function χ compactly supported in $(0,1)$. The matrix

$$G_{ij} = \int_0^1 \chi(t) t^{i+j}\,dt \qquad (0 \le i, j \le n)$$

is positive definite: its quadratic form at a nonzero coefficient vector c is $\int \chi(t)(\sum_i c_i t^i)^2 dt > 0$. Indeed χ is positive on some open interval and a nonzero polynomial has only finitely many roots. Hence G is invertible. For c in a sufficiently small neighborhood of zero, the densities

$$\rho_c(t) = \rho(t) + \chi(t) \sum_{i=0}^{n} c_i t^i$$

remain positive on $[0,1]$ and retain the endpoint germs. This uses the positive minimum of ρ and a uniform bound on the finite perturbation. Their moment vectors are $m(\rho\,dt) + Gc$, which contain a neighborhood.

For the closure assertion, take $0 < \delta < 1$ and smooth cutoffs $0 \le \chi_0, \chi_1 \le 1$, each equal to one near its endpoint, supported in the radius-δ ball around that endpoint. Choose $\delta < 1/2$ so each cutoff vanishes near the opposite endpoint. The density

$$\beta_\delta = \delta + (1-\delta)(\chi_0 + \chi_1)$$

belongs to the family in (2). Its mass on $[0,1]$ is at most $\delta + 4\delta = 5\delta$. Every coordinate of its moment vector lies between zero and 5δ, proving convergence to zero. □

Lemma 3. *Every finite nonzero atomless positive measure μ supported on $[0,1]$ has $m(\mu) \in \mathcal{A}_n$.*

Proof. Suppose $y = m(\mu) \notin \mathcal{A}_n$. Since $\mathcal{A}_n$ is open and convex, separation provides a continuous linear functional L such that

$$L(z) < L(y) \qquad (z \in \mathcal{A}_n). \tag{3}$$

The closure assertion gives $0 \le L(y)$. The functional is nonzero, since otherwise (3) contradicts nonemptiness.

We claim $L(p_n(t)) \le 0$ for every $t \in [0,1]$. If it is positive at an interior point, continuity supplies a nonnegative smooth bump ψ supported inside $(0,1)$ with

$$\int_0^1 L(p_n(t))\psi(t)\,dt > 0.$$

Every $1 + c\psi$, $c \ge 0$, is an allowed density in (2). Its L-moment tends to $+\infty$ as $c \to \infty$, contradicting (3). Continuity extends the claim to both endpoints.

The polynomial $q(t) = -L(p_n(t))$ is nonnegative on $[0,1]$ and nonzero: its coefficients are the negatives of L on the coordinate basis. Its zero set is finite, hence μ-null. Since μ has positive mass,

$$0 < \int q\,d\mu = -L(m(\mu)) = -L(y),$$

contradicting $0 \le L(y)$. This proves the lemma, including $n = 0$. □

3 Identification of the endpoint data

Let f witness $b \in W_n$ and set $g = f^{(n)}$. Then g is continuous, nondecreasing and nonconstant, and $g(0) = 0$. Extend it constantly outside $[0,1]$. Its Stieltjes measure $\mu = dg$ is finite, atomless, supported on $[0,1]$, and has positive mass $g(1)$. For every $0 \le j \le n$,

$$b_j = \frac{1}{(n-j)!}\int_0^1 (1-t)^{n-j}\,d\mu(t). \tag{4}$$

For $j = n$, this is $g(1) = \mu([0,1])$. For $k = n - j \ge 1$, Taylor's integral remainder with zero initial jets gives

$$b_j = \int_0^1 \frac{(1-x)^{k-1}}{(k-1)!} g(x)\,dx.$$

Substitute $g(x) = \mu([0,x])$ and interchange the integrals. The inner integral from t to one is $(1-t)^k/k!$, proving (4). No derivative of the original g is required.

Define the linear map $T_n : \mathbb{R}^{n+1} \to \mathbb{R}^{n+1}$ by

$$(T_n y)_j = \sum_{i=0}^{n} \frac{\binom{n-j}{i}(-1)^i}{(n-j)!}\, y_i, \qquad 0 \le j \le n, \tag{5}$$

with $\binom{k}{i} = 0$ for $i > k$. Before reversing the output coordinates, its matrix is lower triangular with diagonal $(-1)^k/k!$. Thus T_n is invertible, and the binomial theorem gives $b = T_n m(\mu)$. By Lemma 3, choose a density ρ as in (2) with the same moments. Set

$$F(x) = \frac{1}{n!}\int_0^x (x-t)^n \rho(t)\,dt \qquad (x \in \mathbb{R}). \tag{6}$$

Repeated differentiation gives $F^{(n+1)} = \rho$ and zero initial jets through n. Equations (4)–(5) give the terminal jets b_j. The endpoint germs of ρ give all higher endpoint conditions in Theorem 1.

Conversely, construction (6) for every density in (2) produces an element of W_n: its nth derivative is strictly increasing. Therefore

$$W_n = T_n(\mathcal{A}_n).$$

Lemma 2 and invertibility of T_n prove openness, completing Theorem 1.

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4 Prior work and formal status

Smooth representation of interior truncated moment functionals is established in di Dio [3], Theorem 2.5 and Corollary 2.7 of arXiv v2; the latter is Corollary 3.1 in the published paper. These results belong to an earlier moment-theoretic framework, not new machinery here. Our direct proof uses standard separation, polynomial root finiteness and smooth bumps. The source-specific content is the atomless Stieltjes identification, the endpoint-one density requirement, and the all-order endpoint assembly. The prior-work comparison is bounded. No absolute priority claim is made.

The current pinned Lean 4.34.1/mathlib `d13f23b7` development has 129 declarations in 18 modules. A fresh isolated replay treats warnings as errors and checks every declaration's type and axioms; only `propext`, `Classical.choice` and `Quot.sound` occur. It formalizes the literal source hypotheses, Stieltjes measure and kernel identity, invertible map (5), all derivatives of the smooth witness, the endpoint baseline and the separation proof of Lemma 3. A positive-definite bump Gram matrix proves openness of the concrete moment image, and positivity-preserving density combinations prove its convexity. The final endpoint

`RepeatedIntegrals.source_conjecture_371768`

is closed: it quantifies over every natural order with no residual convexity, openness, moment-representation or smoothness premises.

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References

[1] M. R. Burke, M. Haris and Madhavendra, *Repeated integrals of increasing functions*, arXiv:2512.02151v1 (2025).

[2] M. R. Burke, *Comonotone approximation and interpolation by entire functions II*, arXiv:2512.23949v1 (2025).

[3] P. J. di Dio, *Absolutely continuous representing measures of moment sequences with an emphasis on the regularity of the densities*, arXiv:2411.03145v2 (2024). Published truncated-moment version: doi:10.1007/s00013-025-02175-1 (2025).